Networks I: Concepts and Algorithms

Thresholds of Taste: Modeling Music Genre Diffusion in Social Networks

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Abstract

This paper examines how music genre preferences spread through social networks using a K-threshold Linear Threshold Model applied to the GEMSEC Deezer dataset for Croatia, Hungary, and Romania. Instead of setting the adoption threshold in advance, we estimate it by matching simulated genre distributions to Spotify streaming data. All three networks exhibit scale-free and small-world properties, with a clear density gradient that is key to the diffusion results. We find best-fit thresholds of $K = 20$, $K = 15$, and $K = 10$, which converge to similar values when scaled by network degree. This suggests that differences between countries are driven mainly by network structure, not cultural variation. Overall, the results support a complex contagion process, where genre adoption requires repeated exposure from multiple neighbors rather than a single interaction. For platform designers and cultural policymakers, this implies that seeding a genre at isolated hubs is unlikely to trigger broad diffusion; it is local exposure within communities that drives adoption at scale.

Contents

1 Introduction

Music taste is one of the most social things about us. The genres a person listens to signal personality and identity almost instantly, and yet preferences rarely form in isolation. They are borrowed from friends, reinforced by repeated exposure, and reinforced by the gradual accumulation of peer influence who all seem to like the same thing. Streaming platforms like Deezer have made this dynamic unusually visible. By combining social friendship graphs with detailed genre preference data across millions of users, they offer a rare chance to study how cultural taste actually spreads through a real network. For the music industry, knowing how a niche genre escapes its origin community and reaches a mainstream audience has direct commercial stakes. For platform designers, it raises an uncomfortable question about whether recommendation algorithms reflect taste or manufacture it. And for anyone interested in social influence more broadly, it speaks to a fundamental issue: is adopting a new music genre the kind of thing one friend can trigger, or does it require sustained pressure from several people before it sticks?

Most prior work on threshold-based diffusion has either used synthetic network models or focused on a single country, making it hard to disentangle the effect of the diffusion mechanism from the effect of network structure. The GEMSEC Deezer dataset, which covers friendship networks and genre preferences in Croatia, Hungary, and Romania simultaneously, is well suited to address this. Surprisingly, despite being widely used for graph embedding and node classification, no study has applied a threshold contagion model to it, and the structural variation across the three countries has never been used as a comparative test bed for diffusion. Our central question is: under what social contagion threshold K does the diffusion of music genre preferences through a friendship network best reproduce observed real-world genre distributions, and does this threshold vary across countries with structurally different networks? Rather than fixing K in advance, we treat it as an unknown and calibrate it by running Monte Carlo simulations of the Linear Threshold Model across a range of values, comparing each outcome to external streaming data. The best-fit K is the main empirical result, and cross-country differences in that threshold, explained through the structural properties documented in Section 4.1, are the main comparative finding.

The paper is organised as follows. Section 2 reviews the literature on network structure, homophily, and threshold diffusion. Section 3 describes the data and methodology. Section 4 presents results across network characterization, homophily analysis, community detection, and the diffusion simulations. Section 5 discusses what the results mean, and Section 6 concludes.

2 Literature Review

Understanding how music genre preferences spread through social networks requires drawing on four bodies of literature: the structural properties of real-world social networks, the role of homophily in shaping who connects to whom, the formal theory of threshold-based diffusion, and prior empirical work on the Deezer dataset. Each thread contributes a necessary component to the analysis: structure determines the substrate over which diffusion unfolds; homophily explains the initial clustering of tastes; threshold models formalize the adoption mechanism; and prior dataset work establishes what remains to be explained.

2.1 Network Structure: Small-World and Scale-Free Properties

Two landmark papers established the vocabulary for analysing real-world network topology. Watts and Strogatz [1998] showed that many social networks occupy a middle ground between ordered lattices and random graphs: a small number of long-range links dramatically reduces average path length while preserving high local clustering, the defining signature of a *small-world* network. High clustering creates dense local neighborhoods where a genre accumulates exposure quickly; short global paths ensure that once a genre escapes a cluster it can reach the rest of the network in few steps. Barabási and Albert [1999] identified a second universal feature: scale-free degree distributions following a power law $P(k) \sim k^{-\gamma}$, arising from preferential attachment. The resulting high-degree hubs accelerate diffusion because they can simultaneously provide the peer exposures required by many neighbors. Real social networks, including the three Deezer networks studied here, display both properties together, motivating their use as the substrate for our threshold-based diffusion model.

2.2 Homophily and the Clustering of Cultural Tastes

Social ties form disproportionately between similar individuals across virtually every measurable dimension, a principle McPherson et al. [2001] termed homophily. The consequence for information flow is a tension between local reinforcement and global reach: a genre already popular in a cluster will easily accumulate peer exposure within it, but bridge ties carrying content between dissimilar groups are systematically underrepresented. Whether genre adoption behaves as *simple* contagion, where one exposure nearly suffices, or *complex* contagion, where repeated reinforcement is required [Centola and Macy, 2007], determines how sensitive diffusion is to this structure. A low calibrated K points toward simple contagion; a high K toward complex contagion that is more dependent on community architecture. Our cross-country calibration exercise is designed to answer this empirical question.

2.3 Diffusion Models: From Threshold Theory to the Linear Threshold Model

Granovetter [1978] established that collective adoption behaviors can be understood through a single mechanism: individuals adopt when the number of prior adopters in their social environment reaches a personal threshold. His key insight was that identical average thresholds can produce radically different aggregate outcomes depending on the shape of the threshold distribution, shifting analytical focus from individual motivations to system-level interdependency. Watts [2002] extended this to network settings, showing that the probability of a global cascade depends jointly on the fraction of low-threshold early adopters and on the degree distribution of the network substrate. Kempe et al. [2003] gave the Linear Threshold Model (LTM) its modern algorithmic form, in which a node becomes active when the cumulative influence of its active neighbors exceeds a threshold. Our application reframes the LTM as a calibration problem rather than an optimization one: instead of fixing K and identifying the optimal seed set, we treat K as the unknown and ask which value best reproduces observed genre distributions. We also adopt an absolute rather than fractional threshold, so that denser networks like Croatia's inherently offer more opportunities to accumulate K exposures, providing a structural explanation for cross-country differences that a fractional specification would obscure.

2.4 Prior Work on the Deezer Dataset and Music Taste Networks

The dataset was introduced by Rozemberczki et al. [2019] as part of the GEMSEC graph embedding framework. The Deezer social networks for Croatia, Hungary, and Romania document friendship ties alongside multi-label genre annotations across 88 categories, representing a static snapshot from November 2017. Prior uses of these data have focused almost exclusively on node classification via graph embeddings; no prior study has modeled genre spread as a threshold contagion process or attempted a cross-country comparison of adoption thresholds. Rentfrow and Gosling [2003] showed that music preferences are among the most socially salient identity markers, suggesting that adoption may require repeated peer contact consistent with complex contagion. Taken together, these threads establish that music taste is socially embedded, that the Deezer networks exhibit the structural properties that make threshold contagion nontrivial, and that an empirical calibration of K across structurally distinct countries remains an open contribution.

3 Data and Methods

3.1 Dataset

We use the GEMSEC Deezer Social Networks dataset [Rozemberczki et al., 2019], comprising undirected friendship graphs and multi-label genre preferences for users in Croatia (HR), Hungary (HU), and Romania (RO). Edges are stored as integer node-pair CSVs; genre annotations are provided as a JSON file mapping each node ID to a list of liked genres, drawn from 88 distinct labels. A user may hold multiple genre labels simultaneously, reflecting the multi-faceted nature of musical taste. Nodes with no genre annotation are retained for structural computations but excluded from genre-based analyses. Table 1 summarises the three networks, which vary substantially in density and provide a natural setting for cross-country comparison.

Table 1: Dataset Summary

Country	Nodes	Edges	Avg. k
Croatia	54,573	498,202	18.26
Hungary	47,538	222,887	9.38
Romania	41,773	125,826	6.02

3.2 Super-Genre Mapping

Many of the 88 raw labels are closely related subgenres (e.g. Indie Rock, Indie Rock/Rock pop, and Hard Rock are all variants of rock). Using all of them in the diffusion model would introduce severe sparsity, since many genres are held by only a small fraction of users, and would make cross-country comparisons harder to interpret. We therefore aggregate them into 10 super-genres based on musical similarity (Table 2). Labels not covered by any super-genre are excluded from diffusion analyses but retained for network characterization.

Table 2: Super-Genre Mapping

Super-genre	Raw labels (selected)
Pop	Pop, International Pop, Indie Pop
Rock	Rock, Indie Rock, Alternative, Hard Rock
Electronic	Electro, Dance, Techno/House, Trance
Hip-Hop/R&B	Rap/Hip Hop, R&B, Contemporary R&B
Metal	Metal
Classical	Classical, Opera, Film Scores
Jazz/Blues	Jazz, Blues, Country Blues
Folk/Country	Folk, Country, Singer & Songwriter, Bluegrass
Latin/World	Latin Music, Reggae, Corridos, Bollywood
Other	Kids, Stories, Comedy

3.3 Network Characterization Metrics

We compute four descriptors per network. Degree distributions are fit to a power law $P(k) \propto k^{-\gamma}$ on a log-log scale [Barabási and Albert, 1999]. The average clustering coefficient $C = \frac{1}{N}\sum_i c_i$ measures the tendency toward local clustering; high C relative to a random graph indicates tightly knit friendship groups. Average path length L is estimated from 500 randomly sampled source nodes, as exact computation on graphs of 40,000–55,000 nodes is computationally prohibitive. Following Watts and Strogatz [1998], the small-world coefficient is $\sigma = (C/C_{\text{rand}})/(L/L_{\text{rand}})$, where $C_{\text{rand}} \approx \bar{k}/N$ and $L_{\text{rand}} \approx \ln N/\ln \bar{k}$ are Erdős–Rényi baselines; $\sigma \gg 1$ indicates that high local clustering coexists with short global paths.

3.4 Homophily Analysis

For each genre we compute Newman's assortativity coefficient [Newman, 2003] using a binary node attribute (listener vs. non-listener). A positive value indicates that listeners are more likely to be connected to one another than chance predicts; a negative value indicates the opposite. This is applied independently for all 88 genres across each country, yielding a per-genre, per-country homophily profile that reveals which tastes are socially concentrated and which are diffuse.

3.5 Community Detection

We apply the Louvain algorithm [Blondel et al., 2008] to the largest connected component of each network, maximising the modularity score Q. Higher Q indicates stronger community boundaries, which is relevant for diffusion since genres may spread freely within communities but struggle to cross between them. We report Q, the number of communities, and average community size per country. To assess whether communities align with genre preferences, we compute the Normalized Mutual Information (NMI) between each node's community label and its dominant genre, where NMI $= 1$ indicates perfect correspondence and NMI $= 0$ indicates no alignment.

3.6 Diffusion Model

We adopt a K-threshold Linear Threshold Model. At $t = 0$, each user is seeded with the super-genres derived from their actual Deezer profile, mapped through the classification in Table 2. Users with no genre annotation are initialised with an empty set. Because the initial state is fixed by the observed data, the simulation is deterministic for a given K. A user adopts genre g when at least K neighbors already possess it, and this adoption is irreversible. The simulation ends when a full round produces no new adoptions. The use of an absolute threshold rather than just a ratio means the same K is proportionally easier to satisfy in Croatia's denser neighborhoods than in Romania's sparser ones, placing the structural cross-country comparison directly into the model.

3.7 Monte Carlo Simulation and K Calibration

We evaluate $K \in \{2, 3, 5, 7, 10, 15, 20\}$ for each country and run the simulation to equilibrium. Because the initialization is deterministic, each value of K produces a unique equilibrium distribution. The best-fit K minimises the mean absolute error (MAE) between the simulated normalized super-genre distribution $\hat{p}$ and the observed genre popularity p for each country, sourced from a Kaggle dataset based on data from the Spotify API.[1] A low best-fit K implies that genres spread with little social pressure, while a high K implies that users require substantial neighbor adoption before joining.

3.8 Robustness Checks

To assess sensitivity to data noise, we apply a 5% node-dropout perturbation at the best-fit K and at $K \pm 2$: for each run, a random 5% of nodes have their genre profiles removed before diffusion begins, simulating missing data. This is repeated 20 times per K value per country. On the one hand, if the standard deviation of the equilibrium super-genre shares across runs remains low, the results are driven by network dynamics and not by the precise initial genre assignment. On the other hand, substantial variation would indicate fragility to data quality.

4 Results

4.1 Network Characterization

Table 3 reports the structural descriptors for all three networks.

Table 3: Network Summary Statistics for the Three Deezer Social Networks

Country	Nodes	Edges	Avg k	C	APL	γ	σ
Croatia	54,573	498,202	18.26	0.137	4.47	2.39	342.5
Hungary	47,538	222,887	9.38	0.116	5.35	2.73	530.1
Romania	41,773	125,826	6.02	0.091	6.29	2.79	595.6

Note. APL is estimated from a 500-node random sample of the largest connected component. σ is computed relative to an Erdős–Rényi random graph of equivalent density.

[1] See https://www.kaggle.com/datasets/jfreyberg/spotify-chart-data.

All three degree distributions follow a power law $P(k) \propto k^{-\gamma}$ with exponents $\gamma \in [2.39, 2.79]$ (log-log $R^2 = 0.85$–0.90), consistent with scale-free social networks (Figure 1). Croatia's heavier tail and higher mean degree ($\bar{k} = 18.3$) reflect a denser hub structure that facilitates long-range diffusion.

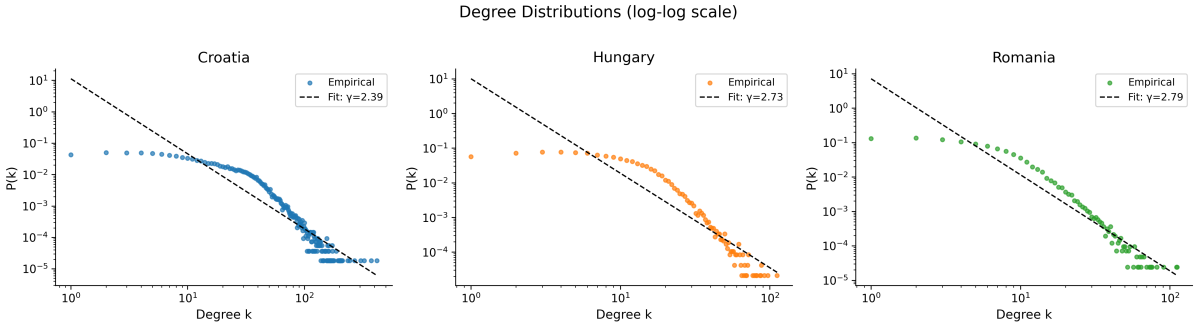

Figure 1: Log-Log Degree Distributions

Note. Dashed lines show power-law fits; fitted exponents γ are reported in the legend.

Clustering coefficients ($C = 0.137$, 0.116, 0.091) are two to three orders of magnitude above the random-graph baseline, while average path lengths remain short (4.47–6.29 hops). This combination yields small-world coefficients $\sigma \gg 1$ for all three countries, confirming that local clustering coexists with efficient global connectivity.

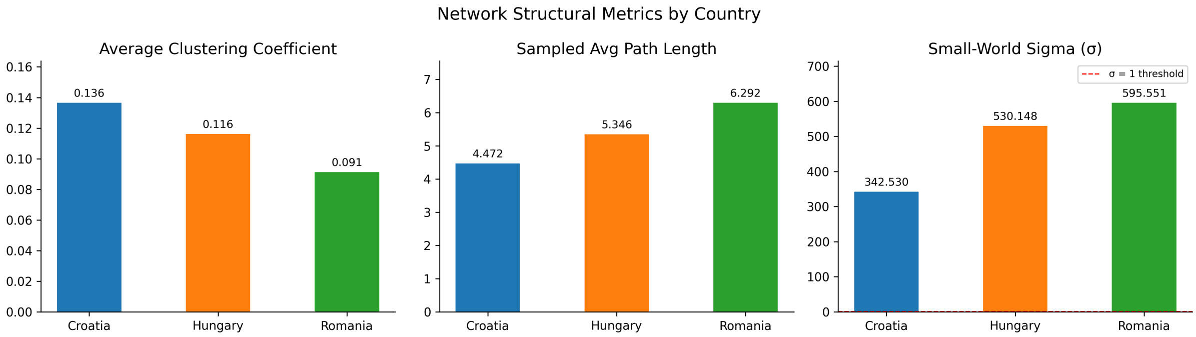

Figure 2: Clustering Coefficient, Average Path Length, and Small-World Sigma by Country

Note. The dashed line marks $\sigma = 1$.

The three networks define a clear structural gradient: Croatia is the densest with the shortest paths; Romania the sparsest with the strongest relative clustering. This gradient bears directly on the diffusion model: because the K-threshold rule counts *absolute* neighbors, the same K is easier to satisfy in Croatia than in Romania, predicting broader genre equilibria in denser networks.

4.2 Homophily Analysis

To quantify the tendency of users with similar tastes to befriend one another, we computed Newman's assortativity coefficient for each of the 88 genres across all three networks, binarising each node as a listener (1) or non-listener (0) of the genre in question. A positive coefficient indicates homophily; a negative one indicates heterophily.

Table 4: Top 5 Most Assortative Genres per Country (Newman's r)

Croatia		Hungary		Romania	
r	Genre	r	Genre	r	Genre
0.062	Jazz	0.057	Rap/Hip-Hop	0.058	Metal
0.040	Blues	0.042	Alternative	0.057	Indie Pop/Folk
0.037	Classical	0.040	Jazz	0.056	Indie Rock
0.034	Indie Rock	0.037	Metal	0.054	Alternative
0.033	Indie Rock/Rock	0.036	Indie Rock	0.053	Indie Rock/Rock

Overall homophily is weak but consistent across all three networks. Across the 80 genres present in all three countries, mean assortativity is low ($\bar{r} = 0.011, 0.011, 0.014$ for Croatia, Hungary, and Romania) yet predominantly positive (79%, 75%, and 71% of genres respectively), indicating that genre-based homophily is a real but modest structural feature of all three networks. Niche genres cluster; mainstream genres do not. Table 4 lists the five most assortative genres per country. Niche tastes such as Jazz, Blues, Metal, Indie Rock, and Alternative dominate the top ranks in every country, whereas mainstream genres such as Pop show near-zero or negative assortativity. This pattern is intuitive: because mainstream genres are liked by almost everyone, genre membership carries little information about friendship structure. Niche genres, by contrast, are shared by a smaller subpopulation whose social ties are more concentrated.

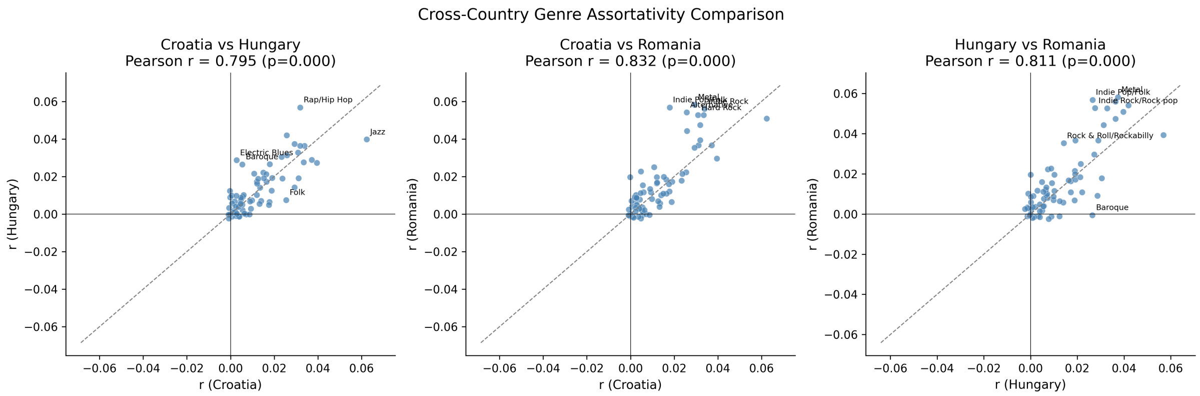

Figure 3: Per-Genre Assortativity Compared Across Country Pairs

Note. Each point is one genre; the dashed line is the diagonal. Annotated points are the five most divergent genres between each pair.

Figure 3 compares assortativity values for the same genre across country pairs. The positive correlations indicate that a genre's homophily rank is broadly stable across countries: a genre that clusters in Croatia tends to cluster in Hungary and Romania too. Romania shows slightly higher assortativity overall, consistent with its sparser, more community-structured network where tighter local groups amplify like-with-like connections.

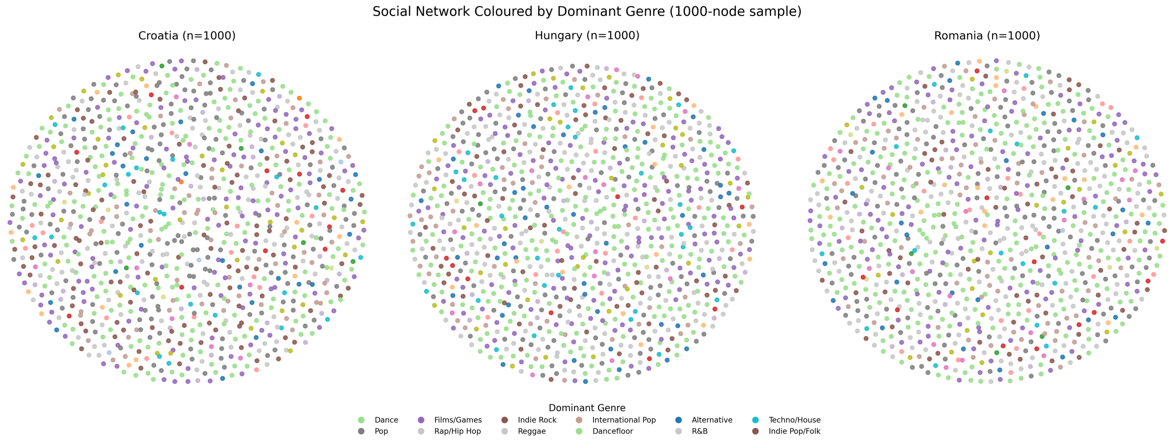

Figure 4: Spring-Layout Visualization of a 1,000-Node Sample from Each Network

Note. Colored by dominant genre. Genre clusters are visible but strongly intermixed.

Figure 4 shows 1,000-node samples of each network colored by dominant genre. Genre clusters are visible but overlapping, consistent with the low assortativity values: there is a tendency for similar tastes to colocate, but the mixing is far from complete. This partial clustering motivates the threshold model, since if genres were fully segregated diffusion dynamics would be trivial, and the interesting behavior arises precisely because users sit at the boundary between taste communities.

4.3 Community Detection

The Louvain algorithm was applied to the largest connected component of each network. Table 5 reports modularity Q, number of communities, average community size, and the normalized mutual information (NMI) between detected communities and dominant genre labels.

Table 5: Community Detection Results for the Three Deezer Networks

Country	**Communities**	**Modularity Q**	**Avg. Size**	**NMI (genre)**
Croatia	24	0.741	2273.9	0.009
Hungary	26	0.678	1828.4	0.007
Romania	54	0.748	773.6	0.013

Note. NMI is computed between Louvain community assignments and dominant genre labels.

All three networks exhibit high modularity, with Q values between 0.678 (Hungary) and 0.748 (Romania), indicating that friendship ties are strongly concentrated within communities rather than distributed across them. The three countries differ markedly in community architecture, however. Croatia produces only 24 communities despite having the largest node count, yielding a mean community size of roughly 2,274 nodes. Romania fragments into 54 communities with a mean size of only 774 nodes, consistent with its lower density and longer path lengths documented in Section 4.1. Hungary sits between the two at 26 communities of average size 1,828. Figure 5 makes these differences concrete. Croatia and Hungary are dominated by a few very large communities, while Romania's size distribution is far flatter. This contrast reflects Romania's more fragmented network structure: without the high-degree hubs that compress Croatia's graph, the Louvain algorithm finds more, smaller natural partitions.

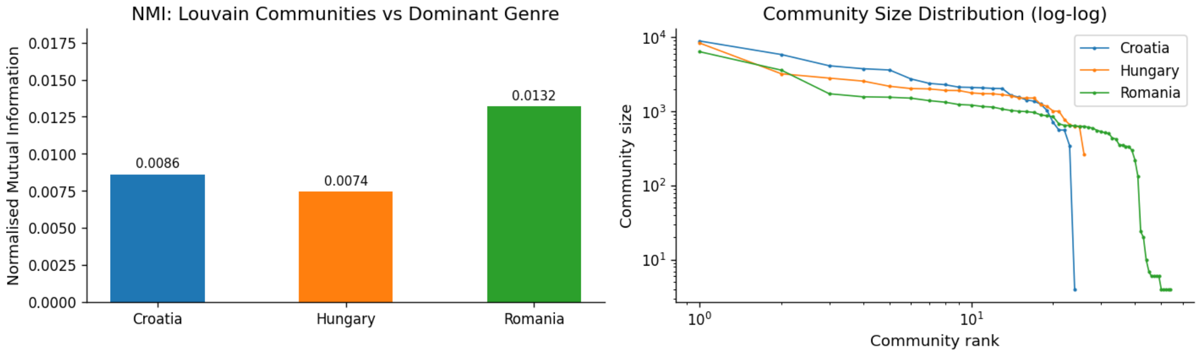

Figure 5: Louvain Communities and Community Size Distribution

Note. Left: NMI between Louvain community assignments and dominant genre labels for each country. Right: ranked community size distributions on a log-log scale.

The left panel reports NMI scores between community membership and dominant genre labels. Scores are uniformly low across all three countries (Croatia 0.009, Hungary 0.007, Romania 0.013), meaning that detected communities do not align closely with genre clusters. Users within the same Louvain community do not systematically share a dominant genre to a degree meaningfully above chance. Romania's slightly higher NMI is consistent with its smaller communities being marginally more taste-homogeneous, but the difference is small. Taken together with the positive but weak per-genre assortativity values in Section 4.2, this confirms that music genre preferences are weakly correlated with social structure rather than tightly segregated by it.

This community architecture has a direct bearing on the diffusion results in Section 4.4. High modularity creates structural bottlenecks at community boundaries: a genre spreading within a community accumulates many within-community exposures but reaches cross-community bridge nodes only rarely. Under a low threshold K, the few inter-community ties are sufficient to carry a genre outward; under a high K, diffusion stalls at the boundary. Romania's combination of high modularity and many small communities makes it the most fragmented landscape of the three, so genres face the most barriers to broad diffusion there. Croatia, with large permeable communities and short path lengths, offers the least resistance. The best-fit K recovered in Section 4.4 should therefore be read not only as an individual adoption parameter but also as a measure of how much peer reinforcement is required to overcome the community structure of each network.

4.4 Monte Carlo Diffusion Simulations

Running the K-threshold Linear Threshold Model across the full grid $K \in \{2, 3, 4, 5, 7, 10, 15, 20\}$ reveals a consistent pattern: for all three countries, lower values of K produce widespread genre adoption, while higher values progressively concentrate adoption around the genres that were already most prevalent in the initial Deezer state. At $K = 2$, for instance, the top three genres in Croatia (Pop, Rock, and Electronic) reach adoption rates of 99%, 98%, and 98% of all nodes, figures that decline to 91%, 69%, and 64% at $K = 20$. The same pattern holds in Hungary and Romania, though at more pronounced rates, as expected in networks with a lower average degree: because fewer neighbors are available to accumulate exposures, the same absolute threshold has a stronger effect.

The calibration results, summarised in Table 6, place the best-fit K at 20 for Croatia, 15 for Hungary, and 10 for Romania.

Table 6: K Calibration Results

Country	**Best-fit** K	**MAE at best** K	**Avg. degree**	K **/ avg. degree**
Croatia	20	0.0093	18.3	1.10
Hungary	15	0.0502	9.4	1.60
Romania	10	0.0624	6.0	1.66

Note. Best-fit K is the value minimizing MAE between the normalized simulated super-genre distribution and the validation target. Croatia's target is its own Deezer observed distribution, while Hungary's and Romania's are from Spotify Charts 2019 (data obtained from Kaggle).

The last column of Table 6 is the most informative result of the calibration. Expressed as a fraction of each network's average degree, the best-fit K is lowest for Croatia (1.09) and highest for Romania (1.67), with Hungary much closer to the latter (1.60). This ordering is the inverse of the density gradient documented in Section 4.1 and is just what the specification of the absolute threshold predicts: because Croatian users have, on average, 18.3 friends compared to 6.0 in Romania, the same absolute number of required exposures is proportionally easier to accumulate there. In other words, the three countries do not need very different social dynamics, they just need roughly the same level of peer pressure. The main difference is how quickly that pressurebuilds up in each country's network. Thus, it is these structural differences and not unique cultural traits that explain why the best-fit K varies between countries.

In Hungary and Romania, the MAE curve (Figure 6) declines steadily and plateaus around the best-fit K. Croatia, lacking external streaming data, uses its own Deezer distribution as the calibration target, so its declining MAE reflects the model converging back toward the initial state, which is an explicit limitation of our project. The Croatia result should therefore be treated as an upper bound, not a precise estimate.

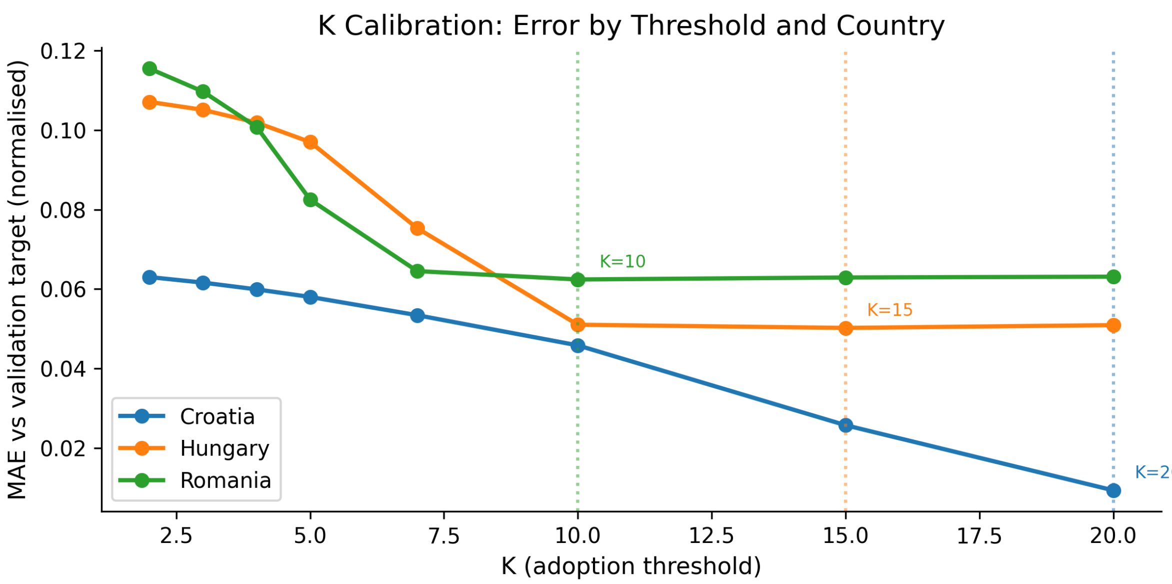

Figure 6: MAE Across K Values per Country

Figure 7 shows the simulated genre distributions at best-fit K alongside the validation targets and the raw Deezer observed shares for each country. The results are significantly better for Hungary than for Romania. In Hungary, the simulation at $K = 15$ broadly tracks the target's ranking, placing Pop clearly ahead of Electronic and giving Hip-Hop/R&B a meaningfully larger share than its raw Deezer proportion alone would suggest. In Romania, however, the simulation at $K = 10$ remains close to the initial Deezer observed distribution and falls short of replicating the Spotify target. Surprisingly, it fails to place Hip-Hop/R&B ahead of Pop, even though the former is the dominant feature of Romania's streaming profile. This suggests that the gap between self-reported genre affiliation in the Deezer data and actual listening behavior as captured by Spotify is widest for Romania, and that no value of K within our tested range is big enough to bridge it. It is unclear whether this is due to a limitation of the threshold model or a mismatch between the two data sources.

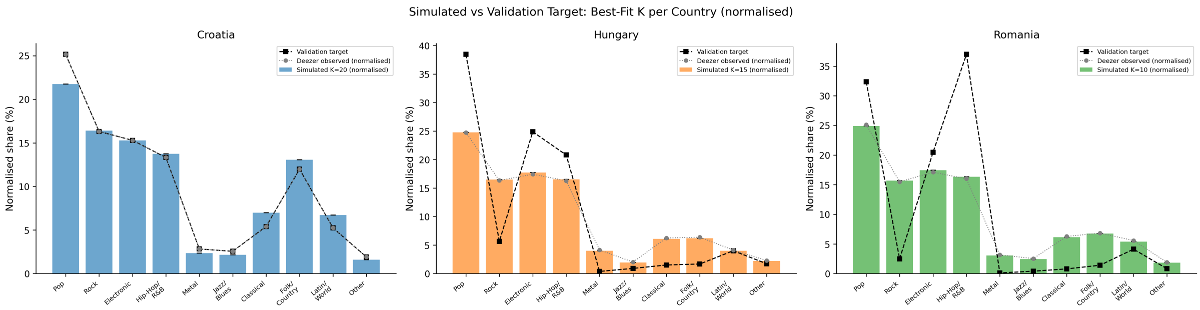

Figure 7: Simulated vs. Validation Target at Best-Fit K

Figure 8 shows the equilibrium distributions under 5% node dropout at the best-fit K and at $K \pm 2$, repeated 20 times per configuration, as a robustness check. In all three countries, the lines for different K values are nearly indistinguishable, and the error bars are minimal. The model's conclusions are therefore not sensitive to small amounts of missing data or to the precise choice of K near the optimum. Thus, we confirm that the network topology and initial genre seeding account for the majority of the effect.

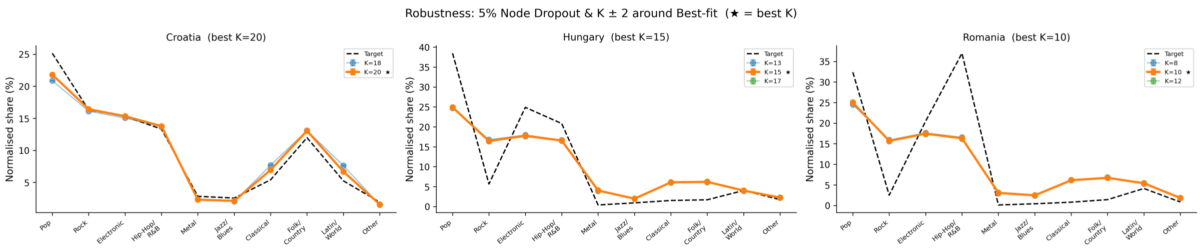

Figure 8: Robustness Check: 5% Node Dropout, $K \pm 2$

5 Discussion

5.1 Interpreting the Best-Fit K

The calibrated thresholds (20 for Croatia, 15 for Hungary, and 10 for Romania) all sit well above 1 relative to each network's average degree, placing music genre adoption within the logic of complex contagion. This means that, under this model, a single friend who listens to a genre is rarely sufficient to trigger adoption: what is required is sustained exposure from multiple neighbors before the behavior sets in. That finding is intuitive in the context of music taste specifically. As Rentfrow and Gosling [2003] established, genre preferences are among the most socially visible identity markers a person carries, which suggests that adopting a new one is not a casual decision. This change, therefore, typically comes from gradual social normalization, not isolated encounters.

It is important to clarify what the calibration establishes and what it does not. For Hungary and Romania, where an external target based on data from Spotify Charts 2019 is available, the best-fit K is a genuine out-of-sample result, that is, the model was not told what the streaming shares looked like, and the threshold that minimizes the distance to those shares is the one we report. For Croatia, however, where no equivalent external data could be obtained and the calibration relies on the Deezer observed distribution itself, the result is in-sample and should be read as an upper bound, not as a precise estimate. The finding for Croatia is still useful for comparing structures, but it is not as strong evidence as the other two.

5.2 Cross-Country Differences

The structural gradient mentioned in Section4.1 (Croatia densest, Romania sparsest, Hungary in between) turns out to be the most relevant feature for understanding why the best-fit K differs across countries. Because the model uses an absolute rather than a fractional threshold, the same K is easier to reach when nodes have more neighbors. Expressed as a fraction of average degree, the best-fit K converges to a ratio of roughly 1.6 to 1.7 in both Hungary and Romania, suggesting that what varies is not how resistant people are to genre adoption, but simply how quickly the required exposures accumulate given each network's density. Thus, differences between countries are driven more by network structure than by cultural factors.

Community structure adds another layer to this picture. Croatia's 24 large communities, combined with its short average path length of 4.47, mean that a genre which escapes its origin cluster can reach the rest of the network quickly and find enough neighbors to keep crossing thresholds. Romania's 54 smaller communities, separated by longer paths, create more structural bottlenecks: a genre accumulating exposures within a community faces a harder crossing each time it reaches a boundary. This is consistent with the observation that, at any given K, the equilibrium genre shares are lower in Romania than in Croatia, as the more fragmented landscape provides fewer opportunities to reach the threshold.

5.3 Homophily and Contagion

The relationship between homophily and diffusion in these networks is more complex than just saying it speeds things up. On the one hand, the niche genres with the highest assortativity (Jazz and Blues in Croatia, Rap/Hip-Hop and Alternative in Hungary, Metal and Indie Rock in Romania) are precisely the genres whose listeners are most socially concentrated. Within their clusters, these genres can accumulate peer exposures quickly, which should give them a head start in crossing local thresholds. On the other hand, the near-zero NMI scores from Section4.3 confirm that genre communities and friendship communities do not align closely, meaning that even the genres with the strongest tendency to connect to similar ones are not restricted to tight pockets. High-assortativity genres are clustered, but not so much that they are isolated.

The tension Centola and Macy [2007] identify is present here, though less strongly. Because the assortativity coefficients are low overall, the friendship network never becomes segregated enough for any genre to be fully contained. Instead of homophily, the main constraint on global spread is the threshold height: with a high K, genres that start with few listeners cannot gather enough exposures from their neighbors to cascade outward, regardless of how sparsely those listeners are distributed.

5.4 Platform and Policy Implications

The calibrated K values have practical implications for how streaming platforms and cultural policymakers think about genre diffusion. If K were low (2 or 3, for instance) a minor algorithmic recommendation suggesting a niche genre to a few well-connected users could trigger a cascade. The high K values we observe here suggest a more demanding picture: for a genre to spread widely, many of a user's friends must have already adopted it before they are likely to follow. This does not mean recommendation algorithms are useless, but seeding a genre through a single highly connected user is unlikely to be enough. A more effective strategy would be to seed multiple users within the same community, building up local adoption before attempting to spread across communities.

For cultural policy, the results point in a similar direction. Romania's high modularity and many small communities mean that niche genres, once established locally, face repeated barriers each time they try to cross the boundary of a community. Policies or platform features that create bridges between disconnected communities, such as collaborative playlists, cross-genre radio features, or socially visible listening activity, could lower the effective threshold for cross-community spread without requiring users to change their individual adoption behavior.

5.5 Limitations

Several limitations of the current analysis deserve to be mentioned. First, the Deezer data is a static snapshot from November 2017, capturing the outcome of diffusion rather than the process itself. The simulation estimates thresholds that could produce this outcome, but it cannot confirm that genres actually spread through the network this way, nor rule out that the distributions mainly reflect the initial seeding.

Second, mapping 88 raw labels into 10 super-genres, while necessary to avoid sparsity, introduces subjectivity. Different aggregation choices could change the distribution of genres and best-fit K values, and sensitivity to this mapping was not assessed. External validation presents a similar challenge: Spotify Charts 2019 reflect actual streaming volume, dominated by popular genres, while Deezer annotations are self-reported and more evenly spread. This discrepancy is most visible in Romania, where the model cannot match Spotify targets regardless of K, highlighting the limits imposed by mismatched data sources.

6 Conclusion

This paper asked under what social contagion threshold K the diffusion of music genre preferences through friendship networks best reproduces observed genre distributions, and whether that threshold differs across countries with structurally distinct networks. Using the GEMSEC Deezer dataset for Croatia, Hungary, and Romania, and calibrating a K-threshold Linear Threshold Model against Spotify streaming data, we find that all three countries are consistent with complex contagion: adopting a music genre requires sustained peer exposure rather than a single contact. The best-fit thresholds are $K = 20$, $K = 15$, and $K = 10$ for Croatia, Hungary, and Romania, respectively. Expressed relative to each network's average degree, these values converge to roughly 1.1–1.7, suggesting that cross-country differences are driven mainly by network density and not by cultural divergences in how people adopt new musical tastes.

Community structure, homophily patterns, and the structural gradient jointly explain both the level of the calibrated thresholds and their variation across countries. The main empirical limitation is Romania, where the simulation cannot match self-reported Deezer affiliation with Spotify streaming behavior, showing that diffusion models depend heavily on the reliability of the data. More generally, the findings ask whether one recommendation can spread cultural tastes, or if it takes gradual and repeated exposure. As we have shown, the evidence favors the latter.

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